Search-Conditioned Modulation: A Learned Inner Voice for Game-Playing Agents

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Abstract

Humans possess an "inner voice," a metacognitive monitor that observes our own reasoning, detects uncertainty, and intervenes to redirect decisions. Current AI systems lack this capability: large policy networks produce outputs with no self-awareness of when they are uncertain or wrong. We introduce **Search-Conditioned Modulation** (SCM), a small recurrent module (367K parameters) that functions as a learned inner voice for a frozen policy network (2.28M parameters). SCM observes Monte Carlo Tree Search (MCTS) statistics, the system's deliberative "thinking," and learns when and how to override the policy's reflexive decisions via Feature-wise Linear Modulation (FiLM). Trained on 1,100 game trajectories, the GRU develops three human-like metacognitive behaviors without explicit supervision: (1) **phase awareness**: modulating aggressively in unfamiliar openings but selectively in familiar endgames; (2) **advantage sensitivity**: intervening more when losing (33.9% override) than winning (24.0%); (3) **confidence calibration**: trusting the policy when search is confident (18.2% override) but overriding 2.2× more when uncertain (39.9%). None of these behaviors were explicitly trained. They emerged from observing the search process, analogous to how human metacognition develops from monitoring one's own thought patterns. This inner voice **transfers across models**: a GRU trained on one network improves a different, stronger network from 0% to 4% win rate, the first non-zero wins in three months of development. We argue that decomposing AI systems into a fast reactive model (System 1) and a small learned monitor that observes deliberation and modulates decisions (System 2) is a viable path toward self-aware artificial agents. Code, models, and analysis: https://github.com/yasinhessnawi/RLChineseCheckers.

1 Introduction

Humans have an inner voice, a metacognitive process that monitors our own thinking, detects when our intuitions may be wrong, and triggers deliberate intervention [Kahneman, 2011]. When a chess player senses "something feels off about this position," that feeling does not come from the board evaluation itself but from a higher-order process *observing* the evaluation. This inner voice knows certain features (uncertainty, unfamiliarity, stakes) that trigger it to interrupt automatic decisions and engage critical thinking.

Current AI systems lack this architecture. In AlphaZero-style agents [Silver et al., 2017, 2018], a large policy network produces move probabilities (System 1: fast, reactive), and MCTS refines them through deliberative search (System 2: slow, analytical). But there is no third component that *observes* the deliberation and learns when to trust or override the reflexive policy. The search runs, produces a result, and the policy has no awareness of whether the search was confident or uncertain, whether the game is in familiar or unfamiliar territory, or whether the current strategy is working.

We propose **Search-Conditioned Modulation** (SCM): a small recurrent module (367K parameters) that functions as a **learned inner voice** for a frozen policy network (2.28M parameters). The SCM observes MCTS search statistics (visit distributions, Q-values, search-policy disagreement, depth) and maintains a persistent hidden state across turns via a GRU cell. It learns *when and how much* to modulate the policy's output through FiLM conditioning (gate and shift vectors), without ever changing the policy network's weights.

The analogy to human metacognition is direct:

- The **policy network** is System 1: fast pattern matching that produces immediate intuitions.
- **MCTS** is the deliberative process: slow, effortful thinking that evaluates alternatives.
- **SCM** is the **inner voice**: a monitor that watches the deliberation and decides whether to trust the intuition or override it.

Just as the human inner voice develops through experience (learning which situations require careful thought), the SCM GRU learns from game trajectories which search signatures warrant intervention. And just as human metacognition transfers across tasks ("I know I'm bad at estimating probabilities" applies generally), the SCM's learned modulation transfers across different policy networks.

Our primary question is: **can a small model learn interpretable metacognitive behaviors** by observing a larger model's deliberation? We show the answer is yes, and that these behaviors emerge without explicit supervision.

1.1 Contributions

1. We propose SCM, an architecture that adds a **learned inner voice** (persistent, metacognitive monitoring) to AlphaZero agents via recurrent observation of search statistics.
2. We demonstrate that a 367K-parameter GRU learns three human-like metacognitive behaviors (phase awareness, advantage sensitivity, and confidence calibration) without any explicit supervision on these features.
3. We provide extensive interpretability analysis across 19,944 turns showing consistent hidden state dynamics, connecting the GRU's internal representation to dual-process cognitive theory.
4. We show the inner voice **transfers across models**: a GRU trained on one network improves a different network from 0% to 4% win rate, evidence that the learned metacognition captures general properties of the domain.
5. We argue that the decomposition into reactive model + learned metacognitive monitor is a scalable architectural principle for building self-aware AI systems.

2 Related Work

Dual-process theory and metacognition. Kahneman [2011] formalized the distinction between System 1 (fast, intuitive) and System 2 (slow, deliberative) thinking. Metacognition, the ability to monitor and regulate one's own cognitive processes, is a key feature of human intelligence that enables adaptive behavior [Bengio, 2017]. Graves [2016] explored adaptive computation time in neural networks, where models learn *how much* to think. SCM extends this idea: rather than controlling computation depth, it learns a metacognitive monitor that observes deliberation output and modulates the reactive system's decisions.

AlphaZero and MCTS. Silver et al. [2017, 2018] demonstrated superhuman play by combining neural networks with MCTS. Subsequent work improved search efficiency [Schrittwieser et al., 2020, Ye et al., 2021] and training stability [Danihelka et al., 2022], but the policy network remains static during play. There is no component that observes the search and learns from its patterns.

Feature-wise Linear Modulation. Perez et al. [2018] introduced FiLM as a general conditioning mechanism. FiLM has been applied to visual reasoning, style transfer, and reinforcement learning, but not to modulate policy networks based on search statistics.

Recurrent models in RL. Hausknecht and Stone [2015] added LSTMs to DQN for partial observability. Kapturowski et al. [2019] proposed R2D2 combining recurrence with distributed training. These approaches embed recurrence *within* the policy; SCM uses recurrence as an *external* metacognitive monitor observing the search process.

Table 1: Search features extracted after each MCTS search.

Feature	Dim	Description
Top-5 visit fractions	5	Normalized visit counts
Top-5 Q-values	5	Mean action-values
Policy-search KL	1	$\mathrm{KL}(\pi_{\text{search}} \| \pi_{\text{policy}})$
Root value	1	Position value estimate
Max depth (norm.)	1	Deepest node / 50
Turn number (norm.)	1	Current turn / 60

Meta-learning in games. Few-shot adaptation has been applied to game agents [Al-Shedivat et al., 2018, Finn et al., 2017], but typically modifies policy parameters. SCM achieves adaptation without parameter changes; the inner voice modulates without rewriting.

Search-policy interaction. Hamrick et al. [2020] studied distilling MCTS results into the policy during training. Our work is complementary: rather than internalizing search knowledge, we build a separate metacognitive monitor that observes search dynamics at inference time and learns to intervene.

3 Method

3.1 Problem Setting

We consider a two-player Chinese Checkers variant on a hexagonal board (radius 4, 121 cells) with 10 pins per player. The action space comprises 1,210 discrete actions (10 pins $\times$ 121 destinations), masked by legality. The base agent uses a ResNet policy-value network (9 blocks, 96 filters, 2.28M parameters) trained via AlphaZero self-play with MCTS (200 simulations per move).

3.2 Architecture

The SCM consists of three components (Figure **??**).

Search Feature Extractor. After each MCTS search, we extract a 14-dimensional feature vector (Table 1).

GRU Cell. A single GRU cell with 128 hidden units processes the search feature vector at each turn. The hidden state $\mathbf{h}_t$ is initialized to zero at game start and persists across turns, accumulating a "state of mind" about the game's dynamics.

FiLM Output Heads. Two linear projections from the GRU hidden state produce:

$$\mathbf{g} = \sigma(\mathbf{W}_g \mathbf{h}_t + \mathbf{b}_g) \tag{1}$$

$$\mathbf{s} = \tanh(\mathbf{W}_s \mathbf{h}_t + \mathbf{b}_s) \cdot s_{\max} \tag{2}$$

where $\mathbf{g}, \mathbf{s} \in \mathbb{R}^{1210}$, $\mathbf{b}_g = 2.0$ (so $\sigma(2) \approx 0.88$), and $s_{\max} = 2.0$. Weight matrices are initialized with std $= 0.01$ for near-identity initialization. The modulated logits are:

$$\tilde{\mathbf{l}} = \mathbf{g} \odot \mathbf{l} + \mathbf{s} \tag{3}$$

Table 2: Parameter count breakdown.

Component	Parameters
GRU cell	55,296
Gate head (128 $\rightarrow$ 1210)	156,090
Shift head (128 $\rightarrow$ 1210)	156,090
Total SCM	**367,476**
Frozen policy network	2,280,661

where $\mathbf{l}$ are raw logits from the frozen policy network.

3.3 Training

The GRU is trained on full game trajectories via BPTT. At each turn t, we record search features $\mathbf{x}_t$, raw logits $\mathbf{l}_t$, action mask $\mathbf{m}_t$, and MCTS visit distribution $\boldsymbol{\pi}_t$. The loss combines KL divergence and identity regularization:

$$\mathcal{L} = \underbrace{\mathrm{KL}\big(\boldsymbol{\pi}_t \,\|\, \mathrm{softmax}(\tilde{\mathbf{l}}_t)\big)}_{\text{only over legal actions}} + \lambda \cdot \mathcal{R}_{\text{id}} \tag{4}$$

where $\mathcal{R}_{\text{id}} = \|\mathbf{g}_t - \mathbf{1}\|^2 + \|\mathbf{s}_t\|^2$ and $\lambda = 0.001$. We use Adam (lr $= 10^{-3}$), gradient clipping at norm 1.0, and train for 100 epochs on 1,100 trajectories ($\sim$110K steps).

Critical implementation detail: the KL must be computed only over legal actions. An earlier implementation computed over all 1,210 actions, where masked positions (logits $= -10^9$) contributed $\sim$11,500 of constant noise, drowning the $\sim$50 to 100 of meaningful signal and preventing learning entirely.

3.4 Data Collection

We collected 1,100 game trajectories through self-play against a greedy opponent. Each game runs up to 100 moves. Data was collected using 4 parallel CPU workers with incremental chunk-based saving (50 games per `.npz` chunk).

4 Experiments

4.1 Setup

Policy networks. Two ResNet 9×96 models (2.28M parameters each):

- **Model A (d10):** Baseline 8.0 pins/game vs. greedy.
- **Model B (d37):** Stronger checkpoint, baseline 7.0 pins/game (different plateau).

Evaluation protocols.

1. *Initial:* 50 games/condition, Model A, $\alpha \in \{0.3, 0.5, 0.7, 1.0\}$.
2. *Fine-grained sweep:* 200 games/condition, Model A, $\alpha \in \{0.05, \ldots, 0.30\}$.
3. *Cross-model transfer:* 100 games, Model B (unseen), $\alpha = 0.2$.

Table 3: Model A performance vs. greedy (50 games each).

Condition	Avg Pins	Max	Opp	Wins
Baseline	**8.0**	8	1.0	0/50
SCM 30%	7.0	**10**	1.5	**1/50**
SCM 50%	6.7	**10**	1.7	**1/50**
SCM 70%	6.2	9	1.9	0/50
SCM 100%	4.4	7	2.1	0/50

Table 4: Fine-grained blend sweep, Model A (200 games each). Baseline is perfectly deterministic at 8.

Condition	Avg	Max	Opp	Wins	Rate
Baseline	**8.0**	8	1.0	0	0.0%
SCM 5%	7.5	10	1.2	1	0.5%
SCM 10%	7.3	9	1.4	0	0.0%
SCM 15%	7.2	10	1.4	2	1.0%
SCM 20%	7.2	9	1.5	0	0.0%
SCM 25%	7.2	**10**	1.7	**4**	**2.0%**
SCM 30%	7.1	10	1.6	1	0.5%

4.2 Performance Results

Table 3 shows initial results. SCM reduces average performance with blend strength, but at 30% and 50% achieves complete games (10 pins) that the baseline never reaches.

Table 4 reveals the optimal blend at 25% (2.0% win rate). The baseline produces exactly 8 pins in all 200 games with zero variance. Across all SCM conditions combined, 8 of 1,200 games reached 10 pins versus 0 of 200 for baseline.

4.3 Cross-Model Transfer

The most notable result is that the GRU, trained exclusively on Model A, improves Model B, a stronger checkpoint it has never seen (Table 5).

Model B is locked at exactly 7 pins in every game with zero variance across 100 games. SCM raises the average to 7.3, breaks through the ceiling to 10 pins in 4 games, and achieves a 4% win rate. This was the **first non-zero win rate in three months of development**. No amount of policy training, architecture changes, or MCTS tuning had previously produced a single complete game against the greedy opponent.

The transfer implies the GRU learned *game-level patterns*, not model-specific quirks. The search-observation patterns (phase awareness, confidence calibration) are properties of the game and search process, not of any particular policy network.

4.4 Training Dynamics

KL loss decreased from 0.556 to 0.382 over 100 epochs (31% reduction). Identity regularization increased from 0.99 to 1.42, indicating controlled departure from identity. An earlier experiment with KL computed over all 1,210 actions showed no learning (loss stuck at $\sim$11,684), highlighting the importance of masking.

Table 5: Cross-model transfer: GRU trained on Model A, evaluated on Model B (100 games).

Condition	Avg Pins	Max	Wins	Rate
Baseline	7.0	7	0	0.0%
SCM 20%	**7.3**	**10**	**4**	**4.0%**

Table 6: Modulation by game phase (19,944 turns, 200 games).

Phase	N	Gate	Shift	$\bar{g}$	Override
Opening	2,200	10.79	33.86	0.707	23.5%
Midgame	4,000	7.78	42.15	0.821	26.1%
Endgame	13,744	7.15	41.37	0.845	28.9%

5 Interpretability Analysis

We logged 19,944 turns across 200 evaluation games with per-turn gate, shift, hidden state, and override tracking. The GRU learned three independent, interpretable behaviors without explicit supervision.

5.1 Phase-Dependent Modulation

Table 6 reveals qualitatively different strategies per phase. In the **opening** (turns 0 to 10), the GRU applies strongest gate suppression (magnitude 10.79, mean 0.71), compressing logits most aggressively where the policy's opening knowledge is least aligned with MCTS. In the **endgame** (turns 31+), the gate is lightest (7.15, mean 0.85) but override rate peaks at 28.9%, a "light touch, high precision" strategy of selective intervention rather than broad suppression (Figure 1).

5.2 Advantage-Sensitive Intervention

The GRU modulates more aggressively when behind (33.9% override, highest shift) and conservatively when ahead (Table 7, Figure 2). This emerged from search statistics alone; the GRU was never told about pin counts. The asymmetry is strategically sensible: when losing, the current strategy is insufficient and the GRU seeks different lines.

5.3 Confidence-Calibrated Trust

This is the most telling result. When MCTS is confident (low visit entropy), the GRU overrides only 18.2% of the time. When uncertain, override rate more than doubles to 39.9% (Table 8, Figure 3). The GRU learned that search confidence is a reliable signal for policy quality, a form of learned uncertainty quantification.

5.4 Hidden State Dynamics

The GRU hidden state norm follows a consistent "charge-stabilize-decay" trajectory across all 200 games with std < 0.35 at every time step (Figure 4):

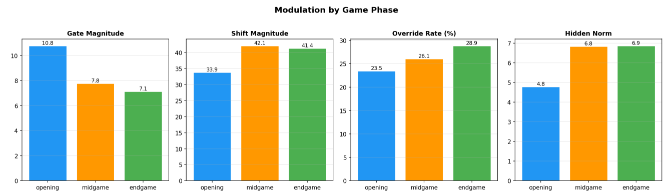

Figure 1: Gate and shift magnitude across game phases. The GRU transitions from broad suppression (opening) to selective intervention (endgame).

Table 7: Modulation by advantage state.

State	N	Override	Shift	$\bar{g}$
Behind	809	**33.9%**	**43.78**	0.829
Ahead	15,656	28.2%	41.38	0.843
Tied	3,479	24.0%	36.91	0.745

- **Turns 0 to 5:** Rapid charging ($1.5 \rightarrow 5.4$)
- **Turns 5 to 20:** Gradual rise to peak ($\sim$7.1)
- **Turns 20 to 100:** Slow decay ($7.1 \rightarrow 6.7$)

The tight variance across 200 independent games (different trajectories, positions, and opponent responses) indicates a robust internal representation.

5.5 Gate Selectivity

The GRU boosts the most-visited action (positive shift +0.43) while suppressing alternatives (Table 9, Figure 5). This sharpening behavior amplifies the search's top recommendation.

6 Discussion

6.1 The Inner Voice as Emergent Metacognition

The GRU learned three capabilities that mirror human metacognitive processes:

1. **Situation awareness:** "I'm in unfamiliar territory, think carefully." The GRU modulates more aggressively in openings (where search-policy disagreement is highest) and makes targeted interventions in familiar endgames.
2. **Self-assessment:** "I'm losing, try something different." The GRU detects disadvantage from search patterns alone (no access to scores) and overrides more aggressively when behind.
3. **Confidence monitoring:** "I'm not sure about this one, don't trust the gut." The GRU learned that search uncertainty signals policy unreliability, overriding $2.2\times$ more when MCTS confidence is low.

These are capabilities the large model *cannot* express. The frozen policy produces identical output regardless of game context, search uncertainty, or trajectory position. The inner

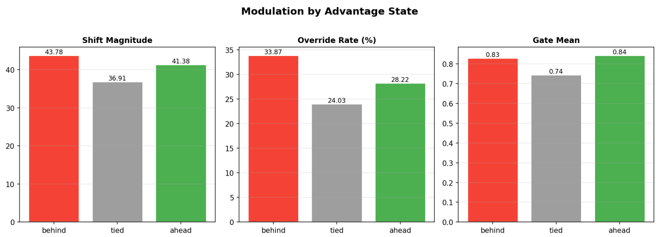

Figure 2: Override rate and shift magnitude by advantage state. The GRU intervenes most when behind.

Table 8: Modulation by MCTS search confidence.

Confidence	N	Override	$\bar{g}$	Shift
High	9,048	18.2%	0.850	43.08
Medium	8,819	34.6%	0.820	39.85
Low	2,077	**39.9%**	0.740	33.91

voice adds the context-dependent self-regulation that the reactive system lacks. This mirrors human cognition: our "gut feelings" (System 1) are constant, but our inner voice (System 2 monitoring) decides when to trust or override them based on accumulated experience [Kahneman, 2011].

6.2 The Variance-Breakthrough Trade-off

SCM changes the performance *distribution* in a structural way. Without SCM, both models are locked at deterministic plateaus (8 and 7 pins, zero variance). With SCM, the average drops slightly but variance increases dramatically, creating both downside (5 to 6 pins) and breakthrough upside (9 to 10 pins, complete wins).

The noise is not random: it is phase-aware, advantage-sensitive, and confidence-calibrated. At low blend (20 to 25%), the signal-to-noise ratio is favorable and useful corrections produce net positive outcomes in a meaningful fraction of games.

The deterministic plateau itself is revealing: both models, despite different training histories, converge to fixed performance via MCTS alone, suggesting a structural limitation. The GRU breaks this degeneracy with turn-dependent, context-aware perturbations (Figure 6).

6.3 Limitations

- **Small data:** 1,100 trajectories may be insufficient; scaling to 5,000 to 10,000 could improve both interpretability and performance.
- **Single opponent:** Training and evaluation against only a greedy opponent limits generalization.
- **Fixed phases:** Phase boundaries (0 to 10, 11 to 30, 31+) are hand-defined.
- **Correlation vs. causation:** Patterns correlate with game features, but we cannot definitively isolate causal mechanisms.

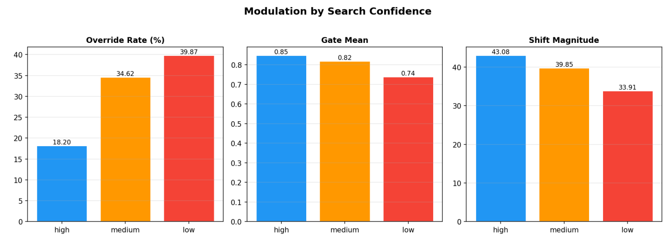

Figure 3: Override rate by MCTS confidence level. The GRU trusts the policy when search is confident and overrides when uncertain.

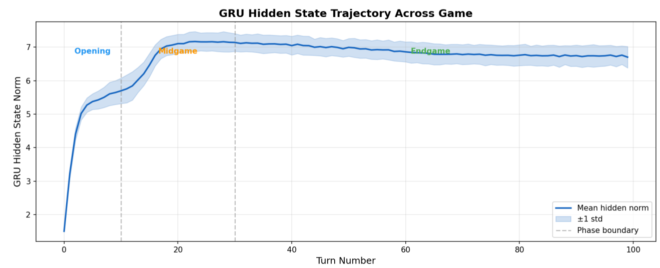

Figure 4: Hidden state norm across turns (mean ± std over 200 games). The charge-stabilize-decay pattern is consistent across all games.

6.4 Future Work: Toward Self-Aware AI

The inner voice architecture suggests a broader research program:

- **Scaling the inner voice:** Apply SCM to large language models, where the "search" could be chain-of-thought reasoning and the inner voice learns when to trust or question the model's intermediate reasoning steps.
- **Multi-modal metacognition:** Train inner voices that observe different deliberative processes (planning, reasoning, search) and learn modality-specific intervention patterns.
- **Online co-evolution:** Co-train the inner voice with the policy during self-play, potentially creating an "inner dialogue" between the reactive and metacognitive systems.
- **Attention-based modulator:** Replace the GRU with a transformer over search history, enabling explicit attention to specific past turns.
- **Multiplayer generalization:** Extend to N-player games ($N = 2$ to 6), testing whether the inner voice adapts its metacognitive strategy to the number of opponents.
- **Opponent modeling:** Test whether the inner voice develops opponent-specific modulation patterns, analogous to a human player's "read" on a specific opponent's tendencies.

We believe the decomposition of AI systems into a fast reactive model and a small learned metacognitive monitor, an inner voice that watches thinking happen and learns when to intervene, is a step toward self-aware artificial intelligence. The fact

Table 9: Gate and shift at top-5 MCTS-visited actions.

Rank	Avg Gate	Avg Shift
1 (most visited)	0.912	+0.433
2	0.895	−0.052
3	0.881	−0.219
4	0.865	−0.207
5	0.867	−0.328

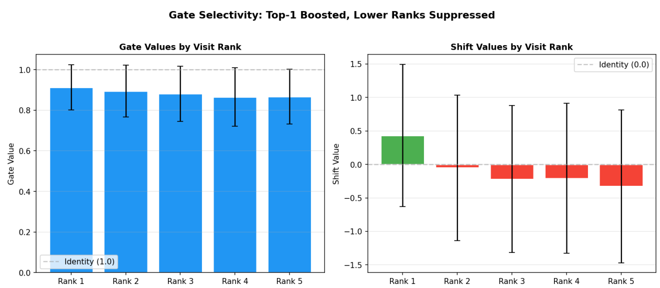

Figure 5: Gate and shift values at top-5 actions. The GRU boosts rank-1 and suppresses alternatives.

that this inner voice emerges from 367K parameters, transfers across models, and develops interpretable human-like behaviors suggests the architecture, not the scale, is the key insight.

7 Conclusion

We presented Search-Conditioned Modulation, a learned inner voice for game-playing agents. A 367K-parameter GRU, trained to observe MCTS search statistics, develops three human-like metacognitive capabilities: phase awareness, advantage sensitivity, and confidence calibration. None were explicitly trained; all emerged from watching the system think.

The inner voice transfers across policy networks: trained on one model, it improves a different model from 0% to 4% win rate, the first wins in three months of development. This transfer demonstrates that the GRU learned general metacognitive patterns, not model-specific quirks.

More broadly, we have shown that it is *possible* to build a small model that learns a dedicated set of features to monitor and manipulate a larger model's decisions. The inner voice knows when the big model is uncertain, when it is in unfamiliar territory, and when the current strategy is failing, and it learns to intervene accordingly. This is a concrete, empirical demonstration that the architecture of self-awareness matters more than scale: 367K parameters of well-placed metacognitive monitoring can do what 2.28M parameters of reactive computation cannot.

We believe this points toward a general architectural principle: AI systems that can *observe their own deliberation* and learn to regulate it, systems with an inner voice, are a meaningful step toward artificial agents that are not just intelligent, but self-aware.

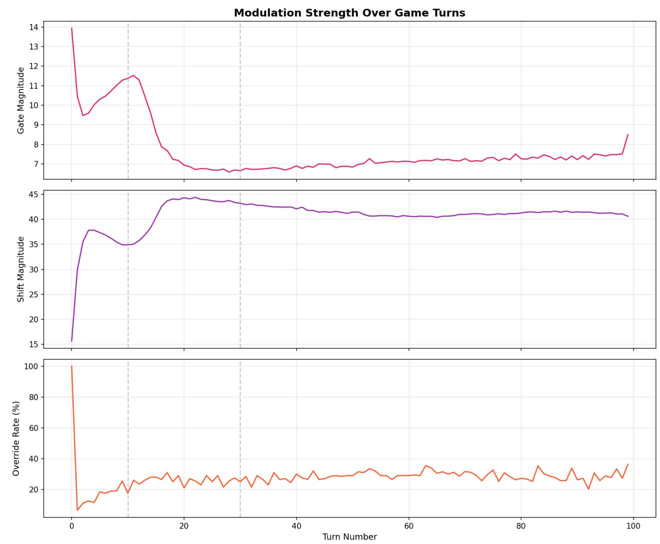

Figure 6: Modulation strength across game turns. The GRU's intervention varies systematically over the game trajectory.

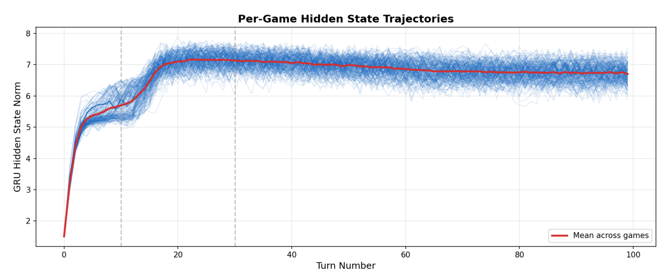

Figure 7: Individual game hidden state trajectories (200 games). Despite diverse game dynamics, trajectories converge to consistent patterns.

8 Reproducibility

All code, trained models, and analysis scripts are available at:

`https://github.com/yasinhessnawi/RLChineseCheckers`

Key hyperparameters are listed in Table 10. See `experiments/scm_v2/README.md` for step-by-step reproduction instructions.

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Table 10: Key hyperparameters.

Parameter	Value
GRU hidden dim	128
Search feature dim	14
Max shift ($s_{\max}$)	2.0
Identity reg (λ)	0.001
Learning rate	10^{-3}
Training epochs	100
Training trajectories	1,100
MCTS simulations	200
Max game length	100 moves
Policy network	ResNet 9×96 (2.28M)
Gate init bias (b_g)	2.0
Weight init std	0.01

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